

# Why does fresh water used by Ai become unusable?

Posted by u/Spongebob\_Despises • 0 points • 19 comments

I've been told all over that ai uses water from fresh water ponds to cool down the machines used to teach the ai. But then they say the water isn't reusable, why can't they re-purify it? Is there no ethical way for ai?

## Comments

u/RazzmatazzUnique6602 • 1 points • 2026-01-17 04:18 UTC

First law of thermodynamics. It shouldn't matter.

u/NombreCurioso1337 • 1 points • 2026-01-16 18:33 UTC

The complaint, so far as I know, is that data centers like to pump fresh clean ground water from wells, depleting (maybe?) the aquifer, and potentially making nearby wells run dry. The water isn't unusable, it just floats away on the wind and will return to earth elsewhere. This isn't an issue with plentiful water, but some places have limited ground water.

They *can* use a closed loop system like nuclear plants, but that would be slightly more expensive for the billion dollar companies, and thus un-American.

u/Friendly\_Preference5 • 2 points • 2026-01-16 08:47 UTC

Water is a rivalrous good, which means a unit consumed is then not available for others. It takes some time to that unit to come back and be available again.

u/simcity4000 • 1 points • 2026-01-15 23:33 UTC

When you hear 'water usage' the actual issue isn't the water itself, it's the energy used to sanitise and move it places.

u/Dman1791 • 3 points • 2026-01-15 23:05 UTC

The main issue is that the liquid cooling parts they use aren't (and probably never will be) food safe. The moment the water isn't drinkable, you can't put it back into the municipal supply without processing.

If they're pulling from a pond rather than the municipal supply, then the "only" issues would be any effects of the heat and contaminants (such as leached metals, or coolant additives) on the local ecosystem.

u/rhomboidus • 5 points • 2026-01-15 23:02 UTC

Potable water used \*for anything\* can not return to the potable water system. All water in a municipal system gets used \*once\* and then goes into the wastewater stream.

**u/MrWedge18** • 6 points • 2026-01-15 22:58 UTC

Data centers use evaporative cooling. The heat from the hardware is used to encourage evaporation, and allowing the hotter water vapor escape into the environment is how the heat gets removed.

**u/TheMrCurious** • 1 points • 2026-01-16 22:58 UTC

Couldn't we catch it like in a distillery?

**u/MrWedge18** • 4 points • 2026-01-16 23:03 UTC

The water molecules are carrying the heat we're trying to get rid of. Recapturing the water also means recapturing the heat

**u/archpaw**n • 2 points • 2026-01-15 22:57 UTC

They can reuse it. This is known as closed-loop cooling. Plenty of datacenters do it. But given how little water they use in comparison to growing food, it's not an issue worth worrying about.

**u/simcity4000** • 0 points • 2026-01-15 23:36 UTC

The water used up in data centres is used to generate shrimp Jesuses and deepfake porn. The water we use up to grow food...feeds people.

**u/archpaw**n • 2 points • 2026-01-16 00:16 UTC

You don't need water-heavy food. You could survive if you ate one less cheeseburger and had rice instead. And you can save more water doing that than by not using ChatGPT in a year. Maybe you'd rather have the cheeseburger, but if someone wouldn't, can you blame them for that?

**u/tysnwl** • 1 points • 2026-01-19 17:53 UTC

Yeah sure I could survive off rice with severe malnutrition?

**u/archpaw**n • 1 points • 2026-01-19 19:53 UTC

You're not going to get severe malnutrition from cutting out one cheeseburger a year.

**u/iAmThe\_Plaza** • 8 points • 2026-01-15 22:52 UTC

The water doesn't actually become "unusable" - it just gets heated up from cooling the servers and then gets released back into the environment at a higher temperature

Most data centers do recycle their cooling water in closed loop systems, but some of it does evaporate and need replacing. The real issue is more about the massive amounts needed rather than it being permanently ruined

**u/NombreCurioso1337** • 2 points • 2026-01-16 18:28 UTC

I've never heard of a data center using a closed loop. Certainly not "most."

**u/ClydePossumfoot** • 3 points • 2026-01-16 23:19 UTC

"Most" is a bit ambiguous here. Classical datacenters are not generally using closed-loop systems, they're almost always open-loop air cooled.

A majority (like 60%+) of AI datacenter builds are closed liquid loops with that number growing. Almost all of the new builds for AI by Google, Meta, MSFT, etc are closed loop.

Almost all (near 100%) high-performance computing (HPC) datacenter builds are immersion/direct-to-chip. They can't really even operate on an air-cooled system.

**u/MTDLuke** • 2 points • 2026-01-15 22:52 UTC

It's mixed with coolant additives to make it better as, you know, a coolant

It's expensive to re-purify it and there's no requirements in place to make them do it anyways

**u/Thedeadnite** • 2 points • 2026-01-16 16:02 UTC

No place does that anyways, you can't use potable water and return it as potable water. All municipal water is processed afterwards by the city water treatment plants unless it's just dumped into the environment.

**u/hellshot8** • 2 points • 2026-01-15 22:52 UTC

Ive never heard anyone say its unusable, where did you hear that?

**u/SuccessfulCurrent957** • 1 points • 2026-01-15 22:51 UTC

Maybe the heat used to cool it down makes it unusable?